

Global Environmental Politics

Carlos Felipe Balcazar

Department of Political Science & School of Public Policy
University College London

January, 2026

Last class: climate change and firms

- Strategic problems hamstringing cooperation for climate regulation.
 - Short-sightedness: to care about the future.
 - Inter-temporal inconsistency: to incorporate costs of inaction.
 - Collective action: more feasible in small/homogenous groups.
 - Credibility: actors' commitment to cooperate needs to be credible.
- Changing the incentives structure of can change behavior.
- Poorly designed incentives can backfire.
 - e.g., resource curse; unconditional transfers.
 - ... important to consider strategic adjustment due to incentives.

Last class: climate change and firms

- Strategic problems hamstringing cooperation for climate regulation.
 - Short-sightedness: to care about the future.
 - Inter-temporal inconsistency: to incorporate costs of inaction.
 - Collective action: more feasible in small/homogenous groups.
 - Credibility: actors' commitment to cooperate needs to be credible.
- Changing the incentives structure of can change behavior.
- Poorly designed incentives can backfire.
 - e.g., resource curse; unconditional transfers.
 - ... important to consider strategic adjustment due to incentives.

Last class: climate change and firms

- Strategic problems hamstringing cooperation for climate regulation.
 - Short-sightedness: to care about the future.
 - Inter-temporal inconsistency: to incorporate costs of inaction.
 - **Collective action: more feasible in small/homogenous groups.**
 - Credibility: actors' commitment to cooperate needs to be credible.
- Changing the incentives structure of can change behavior.
- Poorly designed incentives can backfire.
 - e.g., resource curse; unconditional transfers.
 - ... important to consider strategic adjustment due to incentives.

Last class: climate change and firms

- Strategic problems hamstringing cooperation for climate regulation.
 - Short-sightedness: to care about the future.
 - Inter-temporal inconsistency: to incorporate costs of inaction.
 - Collective action: more feasible in small/homogenous groups.
 - **Credibility: actors' commitment to cooperate needs to be credible.**
- Changing the incentives structure of can change behavior.
- Poorly designed incentives can backfire.
 - e.g., resource curse; unconditional transfers.
 - ... important to consider strategic adjustment due to incentives.

Last class: climate change and firms

- Strategic problems hamstringing cooperation for climate regulation.
 - Short-sightedness: to care about the future.
 - Inter-temporal inconsistency: to incorporate costs of inaction.
 - Collective action: more feasible in small/homogenous groups.
 - Credibility: actors' commitment to cooperate needs to be credible.
- Changing the incentives structure of can change behavior.
- Poorly designed incentives can backfire.
 - e.g., resource curse; unconditional transfers.
 - ... important to consider strategic adjustment due to incentives.

Last class: climate change and firms

- Strategic problems hamstringing cooperation for climate regulation.
 - Short-sightedness: to care about the future.
 - Inter-temporal inconsistency: to incorporate costs of inaction.
 - Collective action: more feasible in small/homogenous groups.
 - Credibility: actors' commitment to cooperate needs to be credible.
- Changing the incentives structure of can change behavior.
- Poorly designed incentives can backfire.
 - e.g., resource curse; unconditional transfers.
 - ... important to consider strategic adjustment due to incentives.

The problem of zero sum

- What someone wins, can be another's loss.
 - ... mutual gain is not possible; someone faces a net cost.
 - ... more resources for one group, means less for others.
 - ... implies some sort of competition for finite resources.

The problem of zero sum

- What someone wins, can be another's loss.
 - ... mutual gain is not possible; someone faces a net cost.
 - ... more resources for one group, means less for others.
 - ... implies some sort of competition for finite resources.

The problem of zero sum

- What someone wins, can be another's loss.
 - ... mutual gain is not possible; someone faces a net cost.
 - ... more resources for one group, means less for others.
 - ... implies some sort of competition for finite resources.

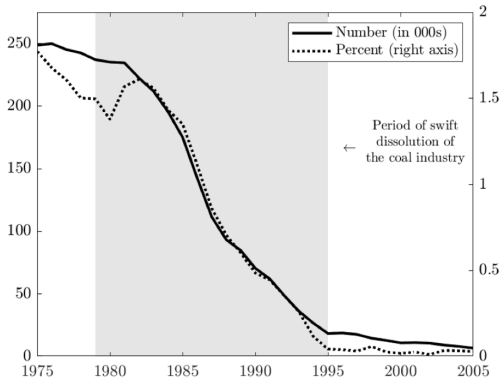
The problem of zero sum

- What someone wins, can be another's loss.
 - ... mutual gain is not possible; someone faces a net cost.
 - ... more resources for one group, means less for others.
 - ... implies some sort of competition for finite resources.
- We need to redistribute the gains, compensating the losers!
 - ... allocations that are Pareto dominant.
 - ... implies *embedded liberalism* or *welfare state*.

The problem of zero sum

- What someone wins, can be another's loss.
 - ... mutual gain is not possible; someone faces a net cost.
 - ... more resources for one group, means less for others.
 - ... implies some sort of competition for finite resources.
- We need to redistribute the gains, compensating the losers!
 - ... allocations that are Pareto dominant.
 - ... implies *embedded liberalism* or *welfare state*.
- Without a social contract, it's hard to sustain cooperation.
 - ... different allocation of resources is seen as a threat.

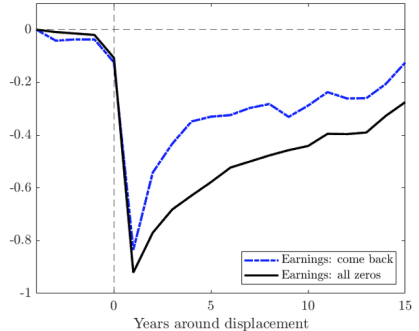
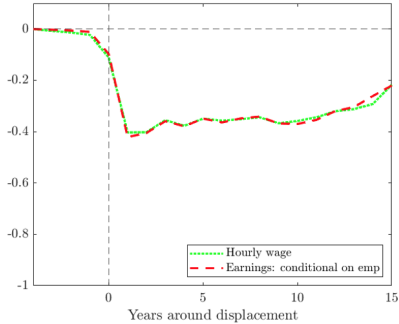
The case of coal



■ Fast transitions can generate negative impacts on communities.

■ These communities exist across the world.

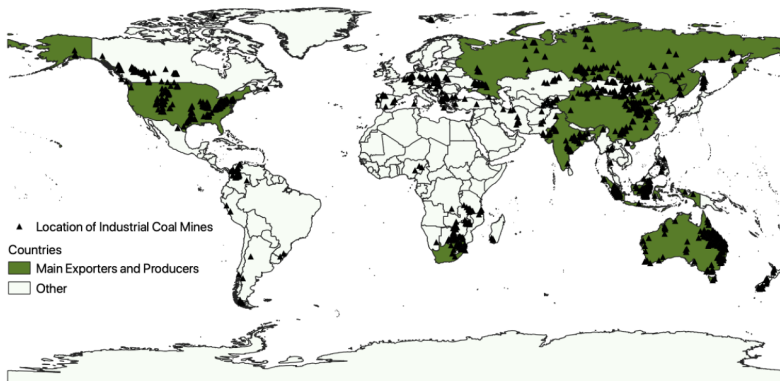
The case of coal



■ Fast transitions can generate negative impacts on communities.

■ These communities exist across the world.

The case of coal



■ Fast transitions can generate negative impacts on communities.

■ These communities exist across the world.

The redistributive effect of adaptation



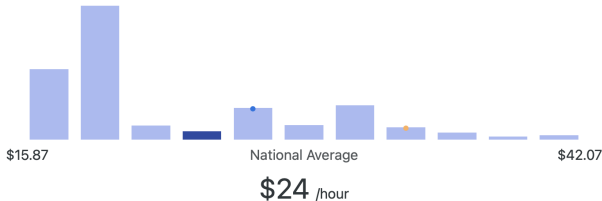
- Adaptation creates winners (e.g., solar) and losers (e.g., coal)!
- Losers oppose transition if transition benefits are low.
- Compensate losers $\Rightarrow \uparrow \text{Pr}(\text{coalitions for climate})?$

The redistributive effect of adaptation

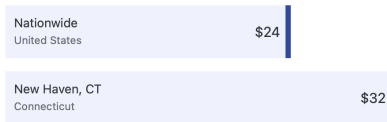


- Adaptation creates winners (e.g., solar) and losers (e.g., coal)!
- Losers oppose transition if transition benefits are low.
- Compensate losers $\Rightarrow \uparrow \Pr(\text{coalitions for climate})?$

The redistributive effect of adaptation

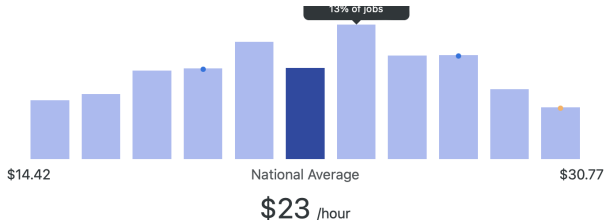


Coal Mine Salary Comparison by Location

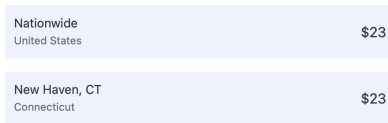


- Adaptation creates winners (e.g., solar) and losers (e.g., coal)!
- Losers oppose transition if transition benefits are low.
- Compensate losers $\Rightarrow \uparrow \Pr(\text{coalitions for climate})?$

The redistributive effect of adaptation

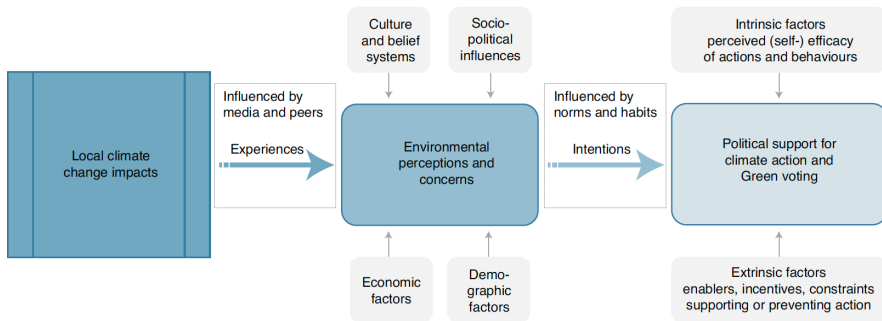


Solar Panel Salary Comparison by Location



- Adaptation creates winners (e.g., solar) and losers (e.g., coal)!
- Losers oppose transition if transition benefits are low.
- Compensate losers $\Rightarrow \uparrow \text{Pr}(\text{coalitions for climate})?$

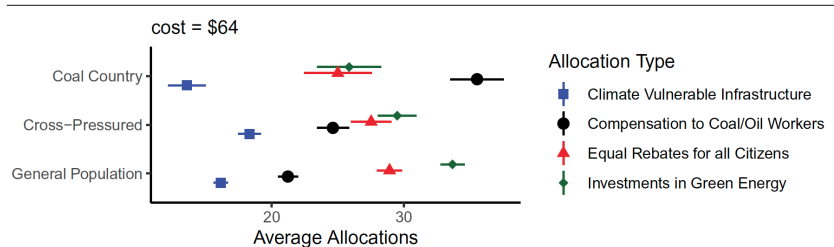
The redistributive effect of adaptation



- Adaptation creates winners (e.g., solar) and losers (e.g., coal)!
- Losers oppose transition if transition benefits are low.
- Compensate losers $\Rightarrow \uparrow \text{Pr}(\text{coalitions for climate})?$

The redistributive effect of adaptation

FIGURE 2. US Preferences for Allocation Purposes of Climate Policy Funds, by Sample

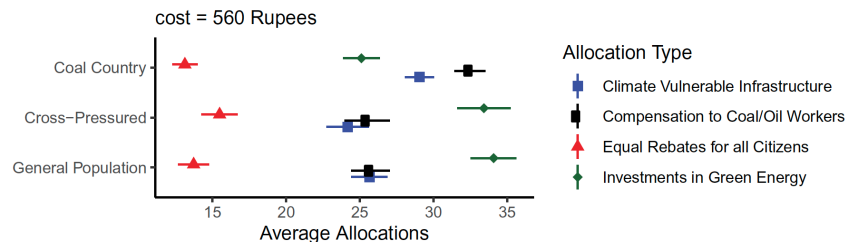


Note: This figure denotes how respondents in our three samples allocated funds raised. *Symbols* represent average allocation, and *lines* represent 95% confidence intervals.

- Adaptation creates winners (e.g., solar) and losers (e.g., coal)!
- Losers oppose transition if transition benefits are low.
- Compensate losers $\Rightarrow \uparrow \text{Pr}(\text{coalitions for climate})?$

The redistributive effect of adaptation

FIGURE 3. India Preferences for Allocation Purposes of Climate Policy Funds, by Sample



Note: This figure denotes how respondents in our three samples allocated funds raised. Symbols represent average allocation, and lines represent 95% confidence intervals

- Adaptation creates winners (e.g., solar) and losers (e.g., coal)!
- Losers oppose transition if transition benefits are low.
- Compensate losers $\Rightarrow \uparrow \text{Pr}(\text{coalitions for climate})?$

More than a money problem



- Not only economic opportunity cost (relative wages).
- Effect on the local social structure (e.g., identity, community).
- Don't interiorize negative externalities on others.
- Economic policies can be myopic; only money incentives.

More than a money problem



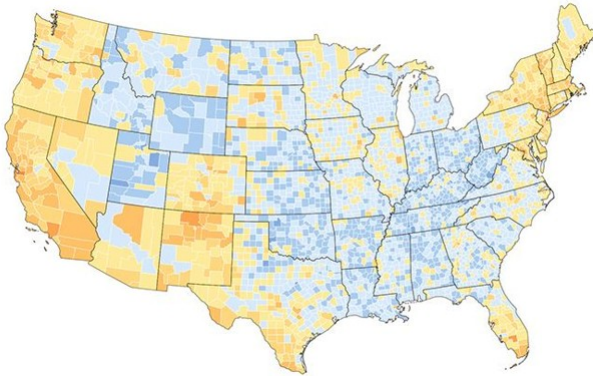
- Not only economic opportunity cost (relative wages).
- Effect on the local social structure (e.g., identity, community).
- Don't interiorize negative externalities on others.
- Economic policies can be myopic; only money incentives.

More than a money problem



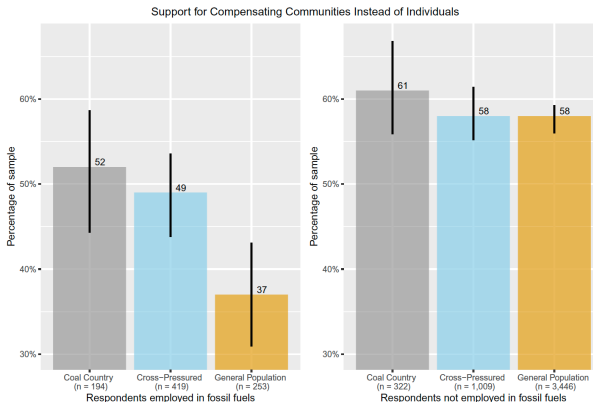
- Not only economic opportunity cost (relative wages).
- Effect on the local social structure (e.g., identity, community).
- Don't interiorize negative externalities on others.
- Economic policies can be myopic; only money incentives.

More than a money problem



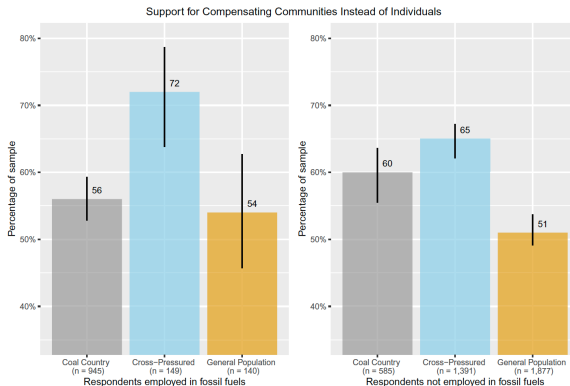
- Not only economic opportunity cost (relative wages).
- Effect on the local social structure (e.g., identity, community).
- **Don't interiorize negative externalities on others.**
- Economic policies can be myopic; only money incentives.

More than a money problem



- Not only economic opportunity cost (relative wages).
- Effect on the local social structure (e.g., identity, community).
- Don't interiorize negative externalities on others.
- Economic policies can be myopic; only money incentives.

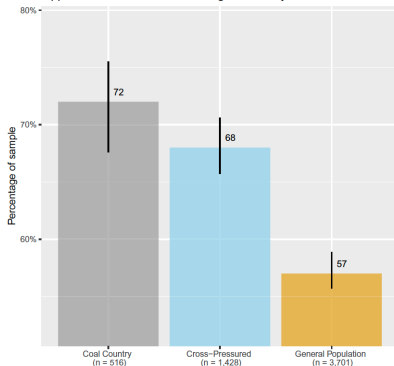
More than a money problem



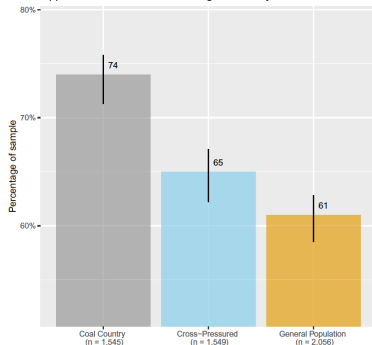
- Not only economic opportunity cost (relative wages).
- Effect on the local social structure (e.g., identity, community).
- Don't interiorize negative externalities on others.
- Economic policies can be myopic; only money incentives.

More than a money problem

Opposition to Policies Threatening the Identity of Coal Communities



Opposition to Policies Threatening the Identity of Coal Communities



- Not only economic opportunity cost (relative wages).
- Effect on the local social structure (e.g., identity, community).
- Don't interiorize negative externalities on others.
- Economic policies can be myopic; only money incentives.

Redistribution is easier said than done

- Programs can have implementation issues; scratched off; lose funding.
- Effective programs should be credible - absence of third party enforcement is limiting.
- Effective programs should be self-enforcing - incentive compatible.
- Hands-tying commitments need to be embedded in the system.
 - ... benefits if alternatives are successful (e.g., shares in companies selling/developing alternative energies).
 - ... generate informal institutions (e.g., identity) around climate cooperation (e.g., morality, economic dependence, etc.).

Redistribution is easier said than done

- Programs can have implementation issues; scratched off; lose funding.
- Effective programs should be credible - absence of third party enforcement is limiting.
- Effective programs should be self-enforcing - incentive compatible.
- Hands-tying commitments need to be embedded in the system.
 - ... benefits if alternatives are successful (e.g., shares in companies selling/developing alternative energies).
 - ... generate informal institutions (e.g., identity) around climate cooperation (e.g., morality, economic dependence, etc.).

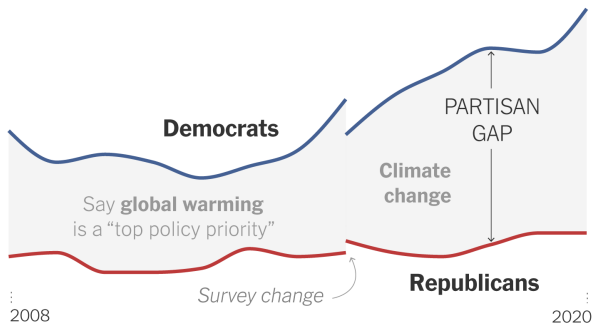
Redistribution is easier said than done

- Programs can have implementation issues; scratched off; lose funding.
- Effective programs should be credible - absence of third party enforcement is limiting.
- Effective programs should be self-enforcing - incentive compatible.
- Hands-tying commitments need to be embedded in the system.
 - ... benefits if alternatives are successful (e.g., shares in companies selling/developing alternative energies).
 - ... generate informal institutions (e.g., identity) around climate cooperation (e.g., morality, economic dependence, etc.).

Redistribution is easier said than done

- Programs can have implementation issues; scratched off; lose funding.
- Effective programs should be credible - absence of third party enforcement is limiting.
- Effective programs should be self-enforcing - incentive compatible.
- Hands-tying commitments need to be embedded in the system.
 - ... benefits if alternatives are successful (e.g., shares in companies selling/developing alternative energies).
 - ... generate informal institutions (e.g., identity) around climate cooperation (e.g., morality, economic dependence, etc.).

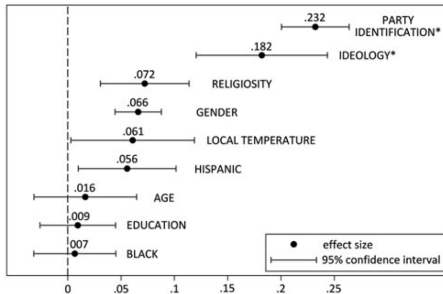
Ideology and other regarding preferences



- Right-wing + religious beliefs $\Rightarrow \uparrow \text{Pr}(\text{climate denialism})$.
- Consistent with strong apprehension towards scientific evidence.
- Citizens may not internalize the negative externalities on others.
- Elites cues can shape attitudes; heuristics to interpret events.

Ideology and other regarding preferences

FIGURE 2 Effects of Variables on Americans' Beliefs about the Evidence for Global Warming

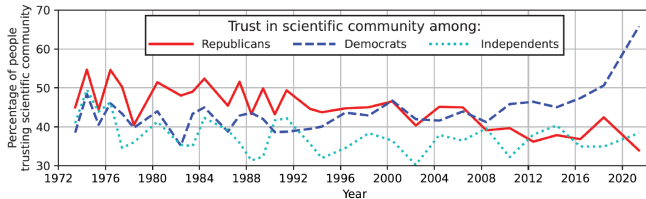


- Right-wing + religious beliefs $\Rightarrow \uparrow \text{Pr}(\text{climate denialism})$.
- Consistent with strong apprehension towards scientific evidence.
- Citizens may not internalize the negative externalities on others.
- Elites cues can shape attitudes; heuristics to interpret events.

Ideology and other regarding preferences

Figure 1

Level of Confidence in Science by Political Party, 1974 – 2022



Source: The General Social Survey, the latest conducted from December 1, 2020–May 3, 2021. Auditors asked, “I am going to name some institutions in this country. As far as the people running these institutions are concerned, would you say you have a great deal of confidence, only some confidence, or hardly any confidence at all in them?” Figure by Alexander Kaurov.

- Right-wing + religious beliefs $\Rightarrow \uparrow \Pr(\text{climate denialism})$.
- Consistent with strong apprehension towards scientific evidence.
- Citizens may not internalize the negative externalities on others.
- Elites cues can shape attitudes; heuristics to interpret events.

Ideology and other regarding preferences

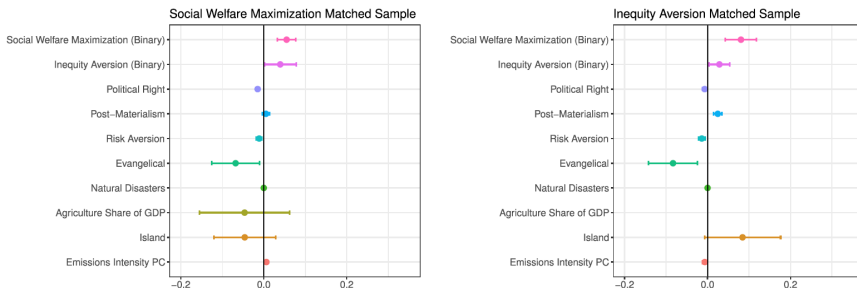
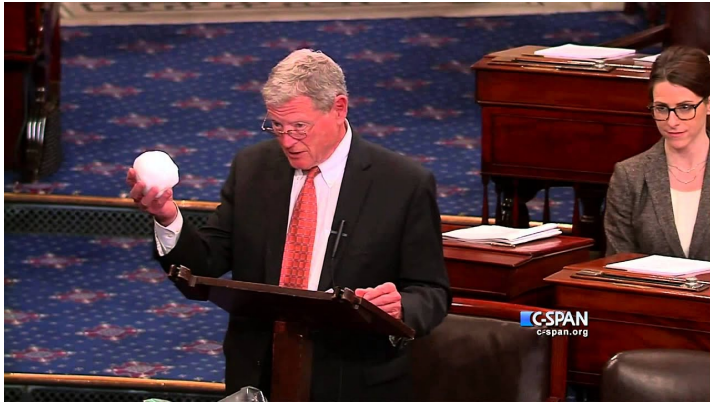


Fig. 2 Marginal Effects of Main Independent Variables. Marginal effects of social preference variables

- Right-wing + religious beliefs $\Rightarrow \uparrow \Pr(\text{climate denialism})$.
- Consistent with strong apprehension towards scientific evidence.
- Citizens may not internalize the negative externalities on others.
- Elites cues can shape attitudes; heuristics to interpret events.

Ideology and other regarding preferences



- Right-wing + religious beliefs $\Rightarrow \uparrow \Pr(\text{climate denialism})$.
- Consistent with strong apprehension towards scientific evidence.
- Citizens may not internalize the negative externalities on others.
- Elites cues can shape attitudes; heuristics to interpret events.

Knowledge/concern for climate change is still low

a

Aware of climate change



- Knowledge of climate change is low in developing world!
- Concern for climate change is low in the developed world!
- Unequal impacts creates costs for cooperation/redistribution.

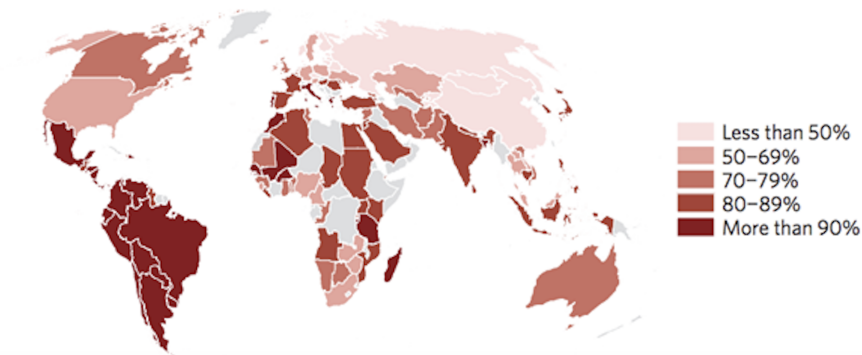
Knowledge/concern for climate change is still low

Rank (of 119)	Country	Awareness (%)	Rank (of 119)	Country	Awareness (%)
1	Japan	98.9%	110	Morocco	30.1%
2	United States	97.7%	111	Togo	29.6%
3	Finland	97.6%	112	Nigeria	27.8%
4	Norway	97.5%	113	Zambia	26.5%
5	United Kingdom	97.4%	114	Ghana	26.4%
6	Australia	97.3%	115	Afghanistan	25.4%
7	Sweden	96.1%	116	Egypt	25.0%
8	Germany	96.0%	117	Burundi	21.7%
9	Netherlands	95.6%	118	Benin	20.7%
10	Canada	95.4%	119	Liberia	20.6%

- Knowledge of climate change is low in developing world!
- Concern for climate change is low in the developed world!
- Unequal impacts creates costs for cooperation/redistribution.

Knowledge/concern for climate change is still low

b Of the 'Aware': climate change is a serious threat



- Knowledge of climate change is low in developing world!
- **Concern for climate change is low in the developed world!**
- Unequal impacts creates costs for cooperation/redistribution.

Knowledge/concern for climate change is still low

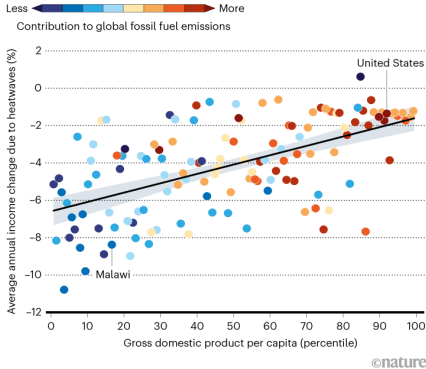
Rank (of 119)	Country	Concern (%)	Rank (of 119)	Country	Concern (%)
1	Ecuador	98.6%	110	Czech Republic	48.1%
2	Bangladesh	98.3%	111	Norway	45.4%
3	Trinidad & Tobago	98.2%	112	Denmark	45.4%
4	Venezuela	98.1%	113	Belarus	44.7%
5	Morocco	97.3%	114	Mongolia	42.9%
6	Costa Rica	96.9%	115	Latvia	42.2%
7	Colombia	96.8%	116	Finland	40.8%
8	Brazil	96.5%	117	Estonia	39.9%
9	Portugal	96.4%	118	Iceland	36.8%
10	Greece	96.3%	119	China	36.1%

- Knowledge of climate change is low in developing world!
- Concern for climate change is low in the developed world!
- Unequal impacts creates costs for cooperation/redistribution.

Knowledge/concern for climate change is still low

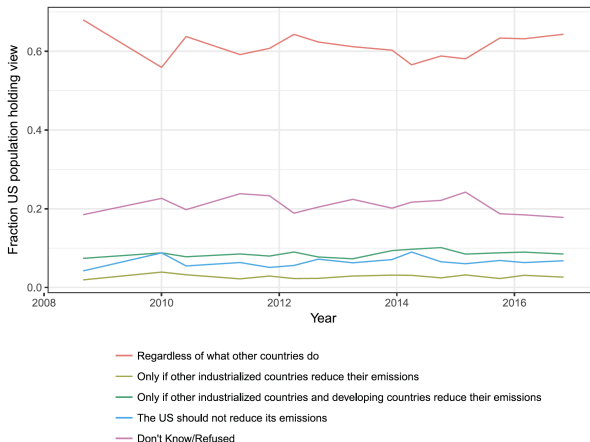
UNEQUAL BURDEN

Despite contributing least to global emissions, tropical and low-income countries suffer the largest economic impacts as a result of heatwaves.



- Knowledge of climate change is low in developing world!
- Concern for climate change is low in the developed world!
- Unequal impacts creates costs for cooperation/redistribution.

Knowledge/concern for climate change is still low



- Knowledge of climate change is low in developing world!
- Concern for climate change is low in the developed world!
- Unequal impacts creates costs for cooperation/redistribution.

Class exercise...

- 1 Make groups of 2 people.



Class exercise...

- 1 Make groups of 2 people.
- 2 Play rock, paper, scissors.



Class exercise...

- 1 Make groups of 2 people.
- 2 Play rock, paper, scissors.
- 3 Write down who losses.



Class exercise...

- 1 Make groups of 2 people.
- 2 Play rock, paper, scissors.
- 3 Write down who losses.
- 4 Those who lost, please wait outside until I call you back.



Five years ago, a country said that it would reduce its use of fossil fuels and work with the UK and other nations on the problem of global warming. In the past five years, the country has increased its use of fossil fuels by a large amount, and it is refusing all efforts to reduce the use of fossil fuels. The country is now encouraging businesses to drill for more fossil fuels. Experts think that the country's use of fossil fuels will double over the next twenty years. The country has high levels of trade with the UK.

Class exercise... cont'

- 1 I will now read a number of alternatives that the UK can use to punish this country.



Class exercise... cont'

- 1 I will now read a number of alternatives that the UK can use to punish this country.
- 2 Raise your hand if you agree with the alternative I mention.



Class exercise... cont'

- 1 I will now read a number of alternatives that the UK can use to punish this country.
- 2 Raise your hand if you agree with the alternative I mention.
- 3 ...but before close your eyes; responses will be anonymous.



Five years ago, a country signed a treaty, in which it promised under international law that it would reduce its use of fossil fuels and work with the UK and other nations on the problem of global warming. In the past five years, the country has increased its use of fossil fuels by a large amount, and it is refusing all efforts to reduce the use of fossil fuels. The country is now encouraging businesses to drill for more fossil fuels. Experts think that the country's use of fossil fuels will double over the next twenty years. The country has high levels of trade with the UK.

Class exercise... cont'

- 1 If you won (lost) in rock, paper, scissors, play with another person who won (lost).



Class exercise... cont'

- 1 If you won (lost) in rock, paper, scissors, play with another person who won (lost).
- 2 Write down who losses.



Class exercise... cont'

- 1 If you won (lost) in rock, paper, scissors, play with another person who won (lost).
- 2 Write down who losses.
- 3 Those who lost, please wait outside until I call you back.



Class exercise... cont'

- 1 If you won (lost) in rock, paper, scissors, play with another person who won (lost).
- 2 Write down who losses.
- 3 Those who lost, please wait outside until I call you back.
- 4 Raise your hand if you agree with the alternative I mention.



Class exercise... cont'

- 1 If you won (lost) in rock, paper, scissors, play with another person who won (lost).
- 2 Write down who losses.
- 3 Those who lost, please wait outside until I call you back.
- 4 Raise your hand if you agree with the alternative I mention.
- 5 ...do not close your eyes this time around!



Class exercise... cont'

- 1 If you won (lost) in rock, paper, scissors, play with another person who won (lost).
- 2 Write down who losses.
- 3 Those who lost, please wait outside until I call you back.
- 4 Raise your hand if you agree with the alternative I mention.
- 5 ...do not close your eyes this time around!
- 6 ...if you change your answer; this will be completely anonymous.

